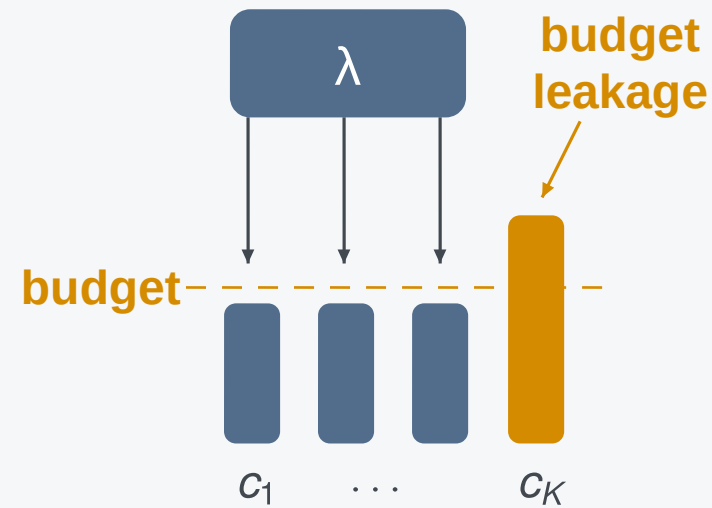


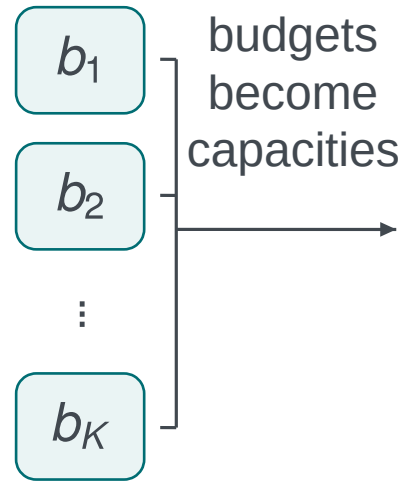
SafeTransport

Penalty-based safe RL

$$\max_{\pi} J_r(\pi) - \lambda^T J_c(\pi)$$



budgets b_k



$S^{(1 \dots K)}$

hard min /
softmin

S

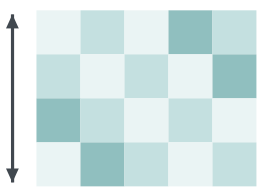
S (capacity ceiling)

y-flow Sinkhorn

$$\min \langle D, F \rangle - \varepsilon H(F)$$

$$0 \leq F \leq S$$

row
scaling



column scaling

$$F^* \leq S$$

F^*

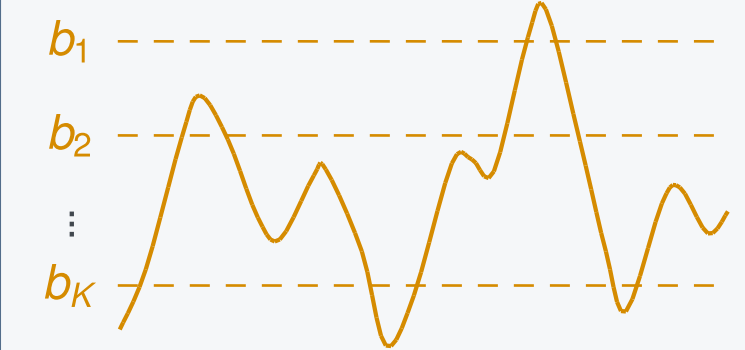
PPO

deployed
policy

deployed-policy
audit

Constraint-based safe RL

$$\max_{\pi} J_r(\pi) \text{ s.t. } J_{c_k}(\pi) \leq b_k, k = 1 \dots K$$



coupled duals oscillate

